Problem 6

Part A: Given complex numbers $z_1 = \cos\frac{t}{2} + i\sin\frac{t}{2}$ and $z_2 = \frac{\sqrt{2}}{2}\sin t + i$. Determine the real parameter t from the condition $z_1 = z_2$.

Part B: Given complex numbers $z_1 = 1 + 2ni$ and $z_2 = 3n - 4i$. Determine the real parameter n such that $\text{Im}(z_1 + z_2) = 0$.

Solution

Part A: Finding t from $z_1 = z_2$

Step 1: Write the Condition

Given:

$$z_1 = \cos\frac{t}{2} + i\sin\frac{t}{2}$$

$$z_2 = \frac{\sqrt{2}}{2}\sin t + i$$

For $z_1 = z_2$, both the real and imaginary parts must be equal.

Step 2: Equate Real Parts

$$\text{Re}(z_1) = \text{Re}(z_2)$$

$$\cos\frac{t}{2} = \frac{\sqrt{2}}{2}\sin t \quad \text{(Equation 1)}$$

Step 3: Equate Imaginary Parts

$$\text{Im}(z_1) = \text{Im}(z_2)$$

$$\sin\frac{t}{2} = 1 \quad \text{(Equation 2)}$$

Step 4: Solve Equation 2

$$\sin\frac{t}{2} = 1$$

This occurs when:

$$\frac{t}{2} = \frac{\pi}{2} + 2\pi k, \quad k \in Z$$

$$t = \pi + 4\pi k, \quad k \in Z$$

For the principal value:

$$t = \pi$$

Step 5: Verify with Equation 1

Substitute $t = \pi$ into Equation 1:

$$\cos\frac{\pi}{2} = \frac{\sqrt{2}}{2}\sin\pi$$

$$0 = \frac{\sqrt{2}}{2} \cdot 0$$

$$0 = 0$$

The equation is satisfied.

Answer for Part A

$$\boxed{t = \pi + 4\pi k, \quad k \in Z}$$

The principal solution is:

$$\boxed{t = \pi}$$

Verification

For $t = \pi$:

$$z_1 = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = 0 + i \cdot 1 = i$$

$$z_2 = \frac{\sqrt{2}}{2}\sin\pi + i = \frac{\sqrt{2}}{2} \cdot 0 + i = i$$

Indeed, $z_1 = z_2 = i$.

Part B: Finding n such that Im$(z_1 + z_2) = 0$

Step 1: Compute $z_1 + z_2$

Given:

$$z_1 = 1 + 2ni$$

$$z_2 = 3n - 4i$$

Adding them:

$$\begin{aligned} z_1 + z_2 &= (1 + 2ni) + (3n - 4i) \\ &= (1 + 3n) + i(2n - 4) \end{aligned}$$

Step 2: Extract the Imaginary Part

$$\mathrm{Im}(z_1 + z_2) = 2n - 4$$

Step 3: Apply the Condition

We require:

$$\mathrm{Im}(z_1 + z_2) = 0$$

$$2n - 4 = 0$$

$$2n = 4$$

$$n = 2$$

Answer for Part B

$$\boxed{n = 2}$$

Verification

For $n = 2$:

$$z_1 = 1 + 2(2)i = 1 + 4i$$

$$z_2 = 3(2) - 4i = 6 - 4i$$

$$z_1 + z_2 = (1 + 4i) + (6 - 4i) = 7 + 0i = 7$$

Indeed, $\mathrm{Im}(z_1 + z_2) = 0$.
The sum is a purely real number.

Final Answer

Part A:

$$\boxed{t = \pi + 4\pi k, \quad k \in Z}$$

Principal value: $t = \pi$
At this value, both z_1 and z_2 equal i.
Part B:

$$\boxed{n = 2}$$

With this value, $z_1 + z_2 = 7$ (purely real).

Summary of Methods

Part A:

- Equate real and imaginary parts separately
- Solve $\sin \frac{t}{2} = 1$ to get $\frac{t}{2} = \frac{\pi}{2} + 2\pi k$
- Verify the solution satisfies both equations

Part B:

- Compute the sum $z_1 + z_2$
- Extract the imaginary part
- Set imaginary part equal to zero and solve for n